

A German Cancer Therapy

By Betty Gwyneth Morales

A non-toxic cancer therapy in Germany draws the usual fire from orthodoxy. A British TV program featured five terminal cancer victims. One of the cancer victims was Peter Newton-Fenbow whose predicted allotted time had run down to two months, maximum. He looked like a living skeleton, had large blotches on his face and body, carried an inoperable tumor in his abdomen, and could barely walk on crutches. Another of the terminal cancer victims was Christina Skinner. She was dying of a metastasized breast cancer and had been told by her doctor that her time had nearly run out too.

Dr. Josef Issell, a German doctor with a clinic in Ringberg, Germany, had been using a new cancer therapy since 1966, discovered by a Bulgarian biochemist, Prodan Christoff, who had experimented successfully on animals. Technically called F-16, the substance has been named CH-23 by its discoverer. Through *Neue Revue*, a German pictorial review, Dr. Josef Issell learned of the five Britains who had appeared on the BBC-TV program as death candidates. Immediately he took the necessary steps to invite the cancer victims to come to his clinic for CH-23 therapy, without charge.

Within four months from the time

that Peter and Christine started CH-23 therapy each was a "living testimonial, a living miracle." Peter, who had been in intense pain for over ten years, went skiing before returning to England.

Christine Skinner has also returned to her family, but her husband, who was apparently healthy, soon died of a sudden heart attack. Now it is doubly important that Christine's life was spared and the once killer cancer has instead apparently been killed by Dr. Issell's timely treatment with CH-23.

Dr. Erich Berger of Klagenfurth, Austria, had a patient 35 years old suffering with advanced cancer of the liver, metastasized to the lymph glands. Her name is Anna Schaefer, and she lives today, because her doctor agreed to try anything which might help to save her life. He sent for Prodan Christoff, the Bulgarian biochemist who discovered and proved CH-23, to aid in administering the new therapy. After just three treatments, one each week, Anna Schaefer's cancerous tumors shrank, and she was able to return home. This was in 1963 and today, after examination by three physicians, there is no sign or symptom of cancer.

In 1964 a 58 year old woman from Italy came to Dr. Issell's clinic because she was suffering from inoperable brain tumor, diagnosed by her physician as fatal. Within a few weeks after her first treatment with CH-23 the tumor could no longer be found, and she was discharged as well, to return home. She is living without having experienced any symptoms, although medical science will not consider her "cured" until after five years without recurrence, which will be 1969.

All of Dr. Issell's patients are also subjected to an intense detoxification program in his clinic at Ringberg, Germany. They are re-educated to a totally nutritious diet of "reformed foods," the German designation for the American equivalent of health foods. For these reasons, as well as for the "unorthodox" CH-23, organized medicine has strenuously opposed Dr. Issell's work, in arrogant disregard of his demonstrated successes with numerous cancer victims counted "expendable" by these same orthodox practitioners.

In spite of the professional pressures, ridicule and intimidation of

other doctors showing interest in CH-23, the clinics in Ringberg and Bad Reichenhall are filled to overflowing. The pilgrimage of once hopeless human wrecks meets those departing, full of new life and renewed hope, and the waiting list grows.

How long will the American public permit the sacrifice of precious human lives, endure this needless suffering? One public official said, in an interview, that a cancer cure would ruin the U.S. economy!

In Germany a "Committee Against Tyranny in Medicine" has been formed. In the U.S. Mrs. Cecile P. Hoffman, herself a former terminal cancer victim, has lived (through Laetrile) and formed "Cancer Victims and Friends" (International), 5525 El Cajon Blvd., San Diego, California. It is non-profit and educational. \$5.00 makes you a member, plus the monthly *Cancer News Journal*.

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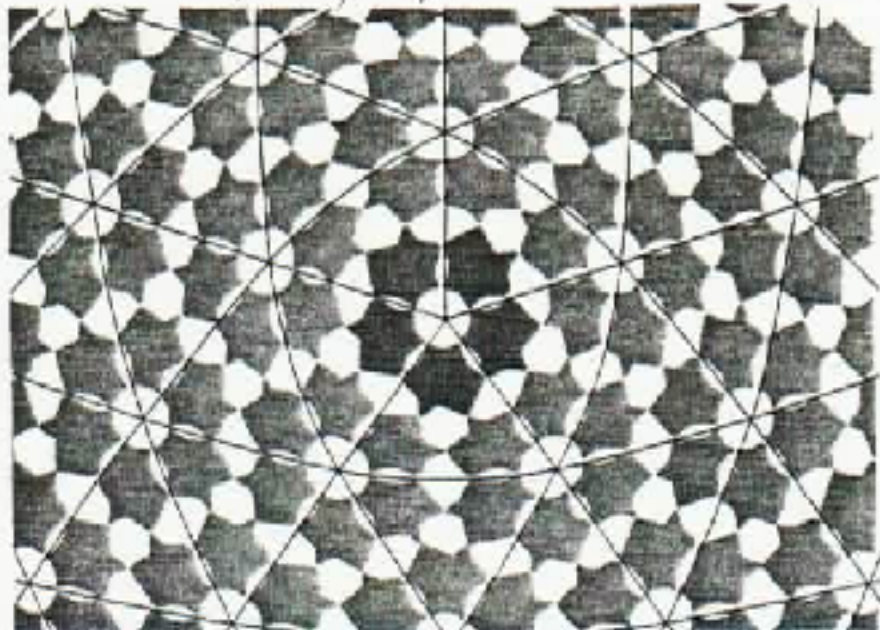
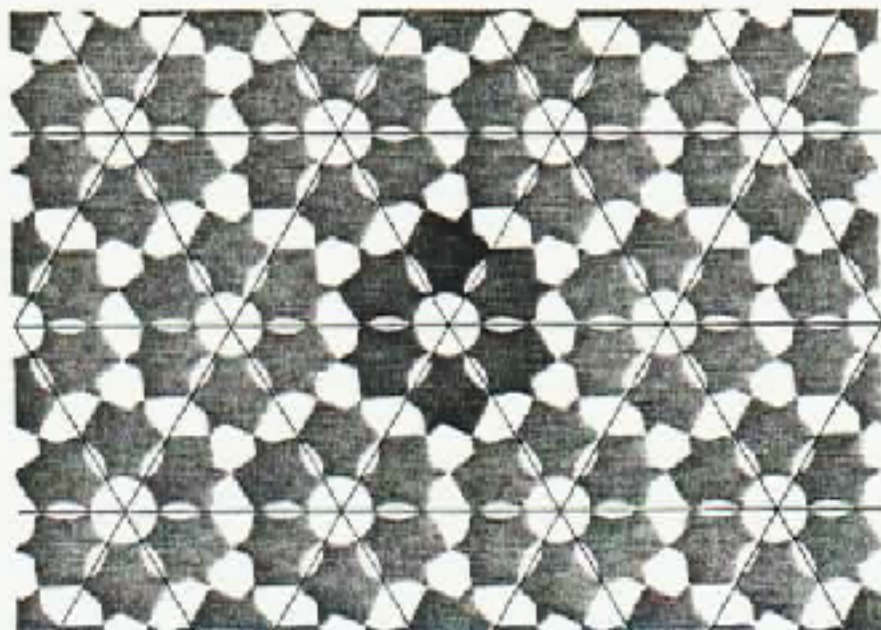
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VIRUS SHELL can be visualized as being folded from a hexagonal lattice of asymmetric shapes (left) in which groups of six shapes (hexamers) can be distinguished (dark shapes). If a 60-degree sector is removed from such a lattice, the edges can be rejoined with-

out disturbing the fit of the shapes; a cone is formed and a hexamer is replaced by a pentamer. This illustration is based on a pattern devised by D. L. D. Caspar of the Children's Cancer Research Foundation and Aaron Klug of the University of Cambridge.

specific function. The absence of gene No. 21 induces the formation of abnormal heads that apparently contain no DNA or only very small amounts of it. Gene No. 31 contributes a "solubility factor," without which the major protein derived from gene No. 23 aggregates into lumps instead of staying in solution and is therefore not available for the assembly of the capsid. Two of my students have demonstrated the action of gene No. 31 both in the test tube and in living cells of the bacterium in-

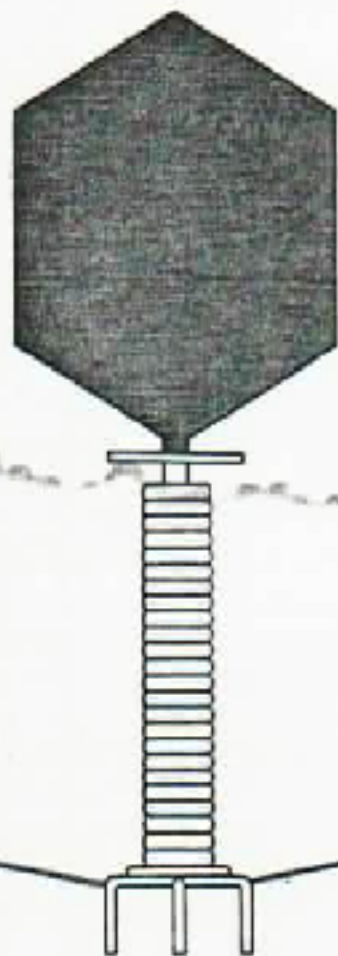
fectected by the T4 virus. If one applies the product of this gene to disrupted cells that contain lumps of the major virus protein, the lumps disappear. This test-tube result is confirmed with a particular mutant virus in which the product of gene No. 31 is inactive when it is formed at a high temperature but is activated as the temperature is lowered. When cells infected with this mutant are incubated at a high temperature, the lumps are produced; as soon as the temperature is lowered the lumps disappear and the

redissolved product of gene No. 23 becomes available for virus assembly.

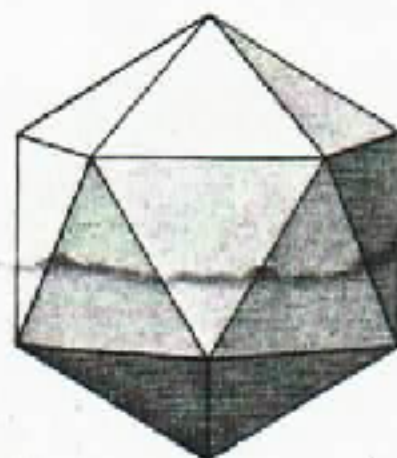
The fact that the same major subunit can be assembled into at least three morphologically distinct particles—the normal head, the short head and the polyhead—shows that the subunit derived from gene No. 23 is not fully shape-specifying. In an effort to understand how the head genes work we have concentrated on the polyhead, which appears to be a very simple variant. First we undertook to find out what order of

Scientific American

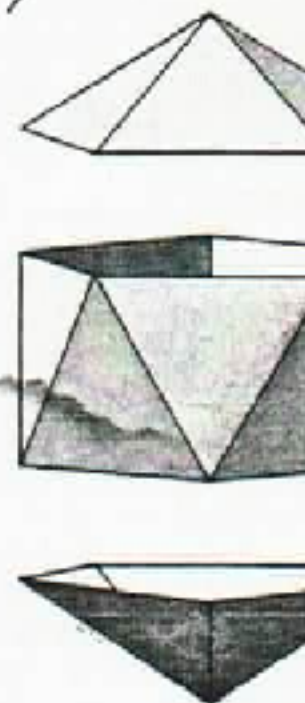
a



b

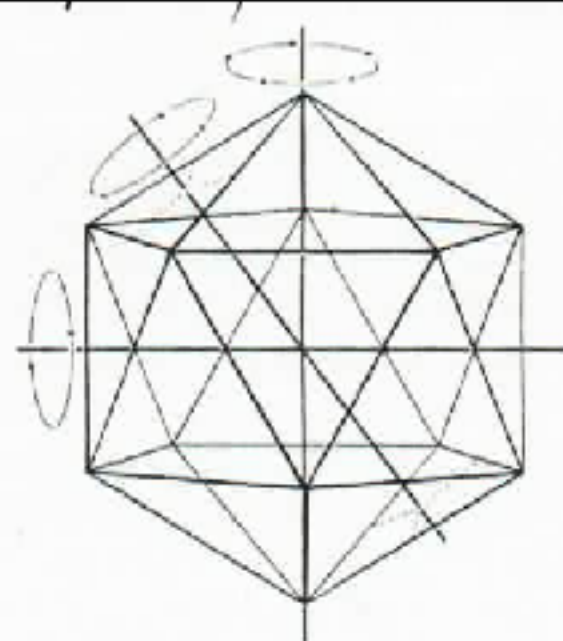
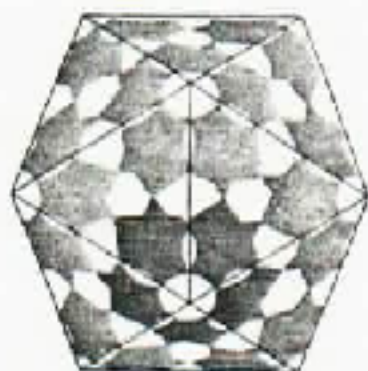


Pyramidal structure for pe



T4 BACTERIOPHAGE is shown schematically (a) as its form has been deduced from micrographs such as the one at the top of

page 33. It now seems that the head capsid is a kind of prolate icosahedron. It is as if two pyramidal vertexes of an icosahedron



ICOSAHEDRON, a polyhedron with 20 faces and 12 vertexes, is built from the hexagonal lattice (*top of opposite page*). If each hexamer is converted into a pentamer, the result is the smallest virus shell for this lattice, a regular icosahedron with 60 subunits

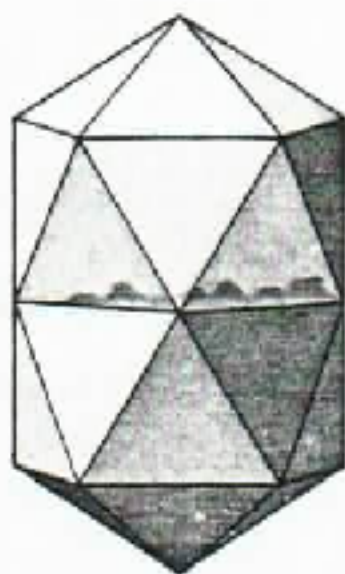
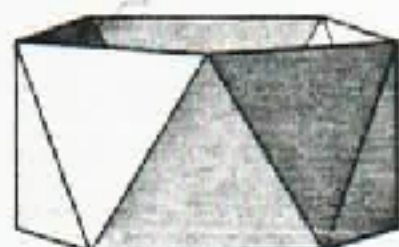
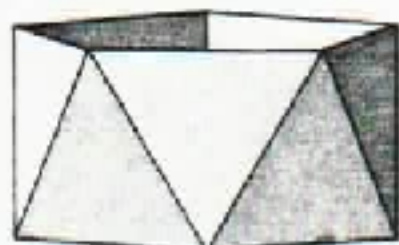
(*left*). The next larger shell for this lattice has 20 hexamers in addition to the pentamers, or 180 subunits (*center*). Like the shell at the left, it has icosahedral symmetry (*right*), that is, it has three kinds of axes of symmetry (*color*): fivefold, threefold and twofold.

morphopoiesis its assembly represents. To do so we made double mutants: viruses in which the functions of two head genes were blocked by mutations. In each case one mutation was in gene No. 20 (the polyhead gene) and the other in one of the other known head genes [see *bottom illustration on next page*]. The results indicate that polyheads are not formed without the active products of at least three genes: No. 23, No. 66 and No. 31. Since No. 31 produces a solubility factor, it is unlikely to contribute

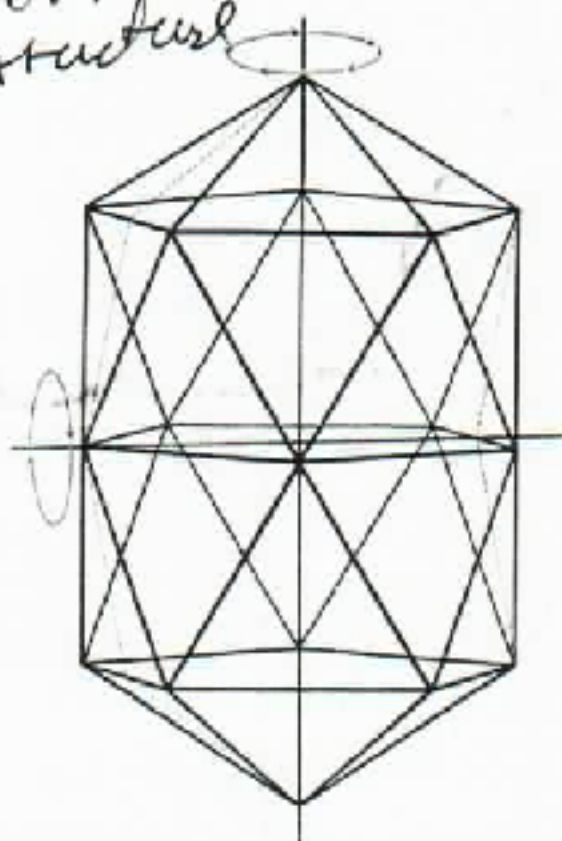
morphopoietic information. The other two, however, are certainly morphopoietic. We can say, then, that the morphopoiesis of polyheads is probably of the second order, with two independent functions carried out by the major protein of gene No. 23 and by the elongating factor of gene No. 66. This comparatively low order of morphopoiesis of the polyheads means that we can hope soon to know in detail just what the factors do; then it should be possible to isolate and purify them, learn their chemistry

and eventually attempt to assemble polyheads in the test tube.

It is already clear that the shape-specifying information cannot be very stringent. This is indicated by the wide variability in the diameter of the polyheads and by the fact that various degrees of pitch can be imparted to the folded hexagonal lattice of subunits that forms a polyhead cylinder. The pitch can be measured because the polyheads in a sample prepared for electron microscopy are flattened and two layers of

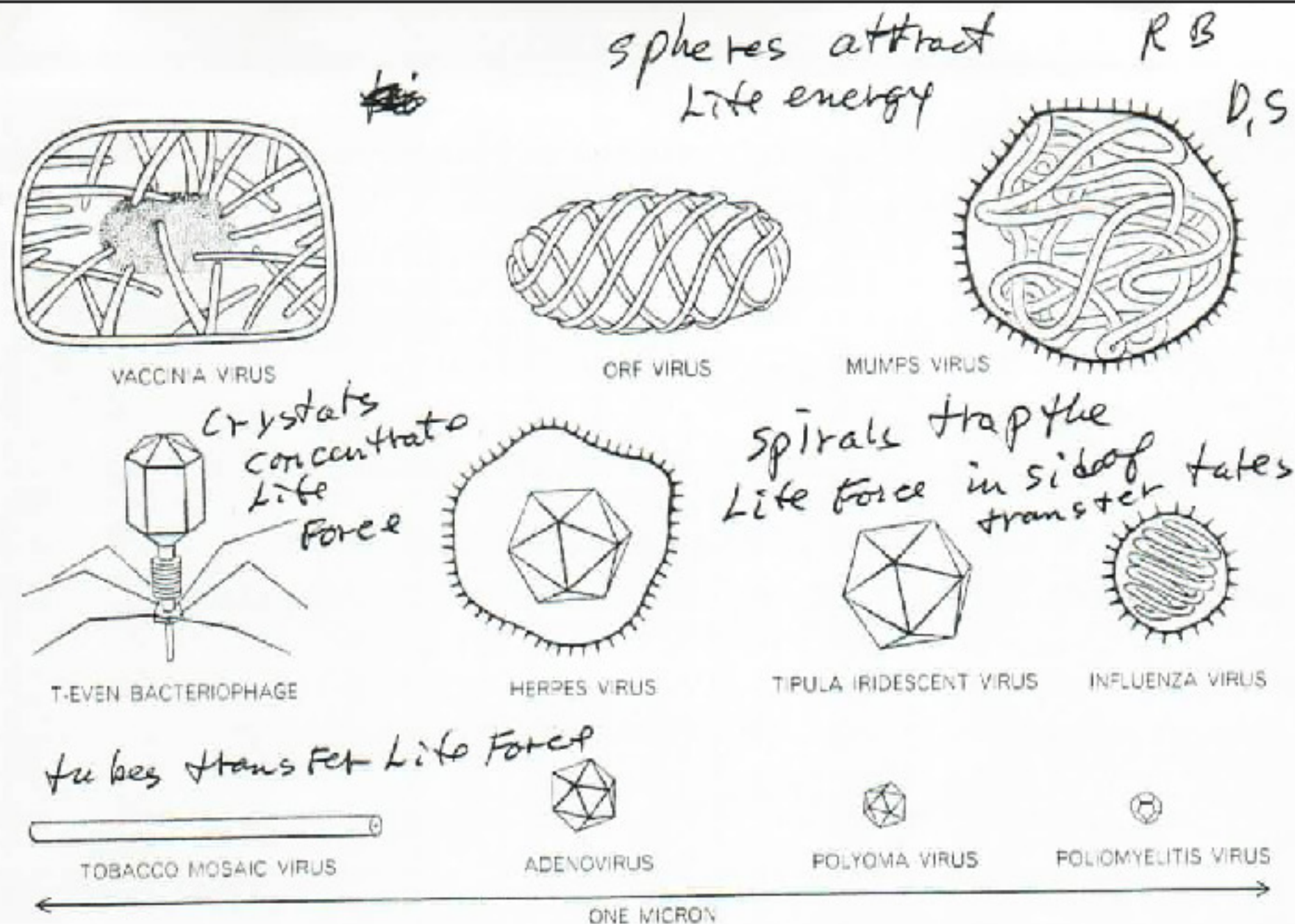


Pyramidal Type structure



were separated by two 10-faceted "equatorial bands" instead of by the usual one band (*four drawings at "b"*). Whereas the icosahedron

has three classes of symmetry, however, the prolate bacteriophage capsid has only two classes: fivefold and twofold (*c*).

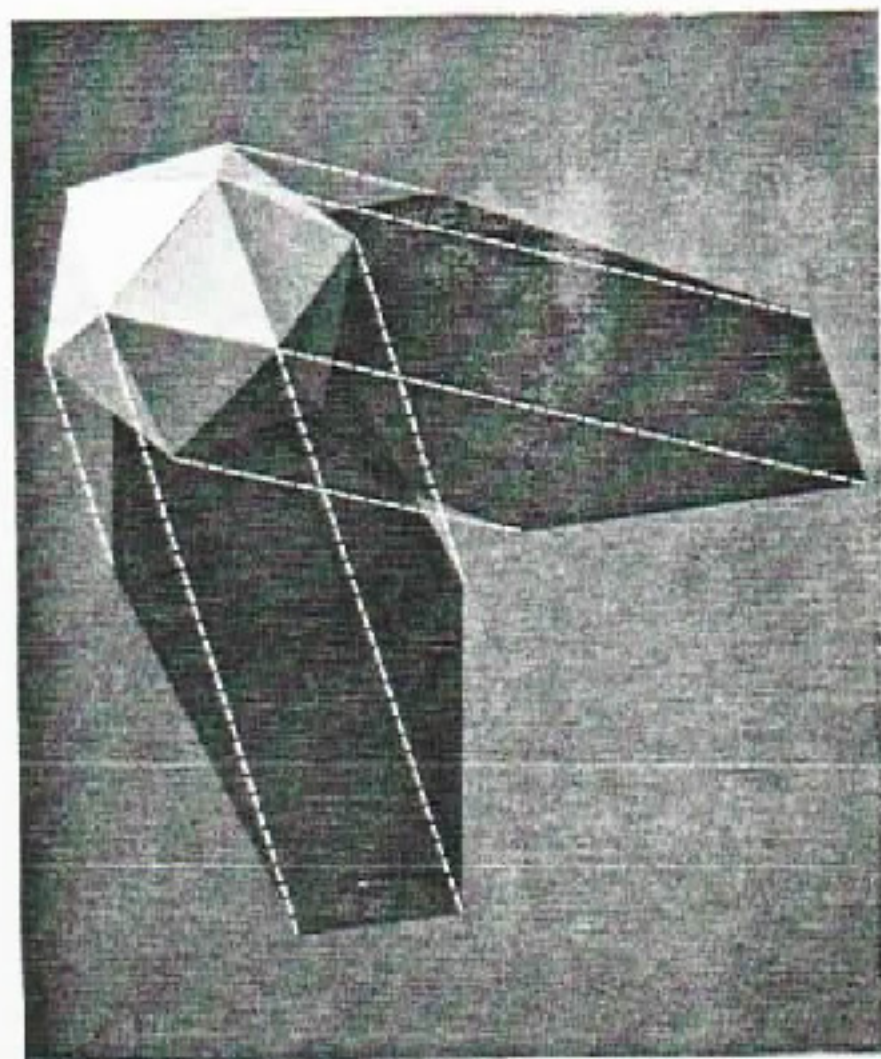


RELATIVE SIZES OF VIRUSES are shown in this chart. A micron, used as a measuring stick, is a thousandth of a millimeter; it is enlarged 175,000 times. The five viruses with polyhedral struc-

tures possess cubic symmetry. The tobacco mosaic virus and the internal components of influenza and mumps virus have helical symmetry. The remaining viruses exhibit complex symmetry.

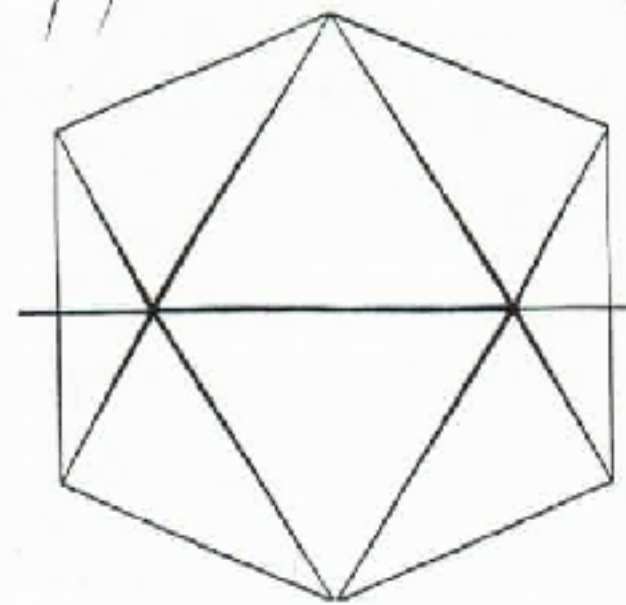
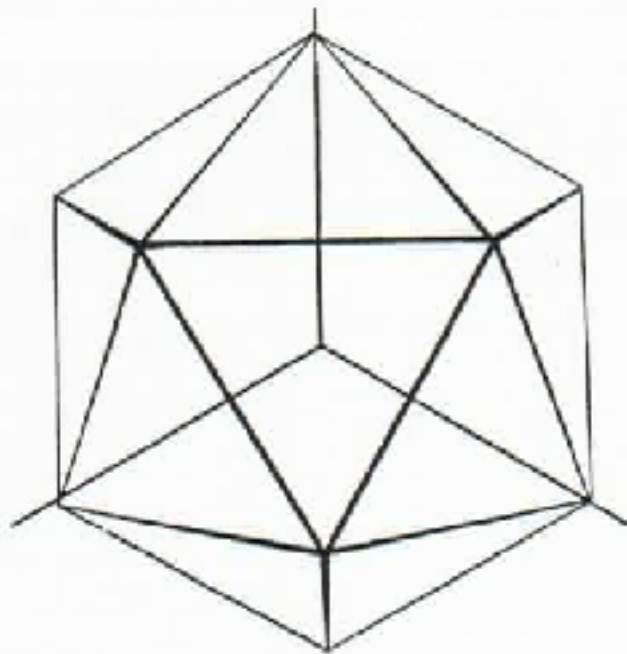
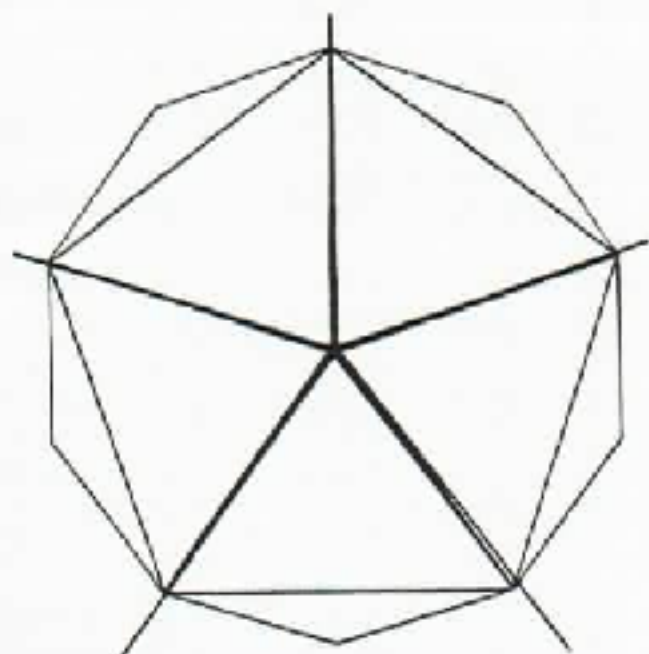


TIPULA IRIDESCENT VIRUS, an insect virus, is so large that its geometrically regular structure shows up clearly when specimens are shadowed with atoms of a heavy metal and enlarged in the electron microscope. In the doubly shadowed micrograph (left) the



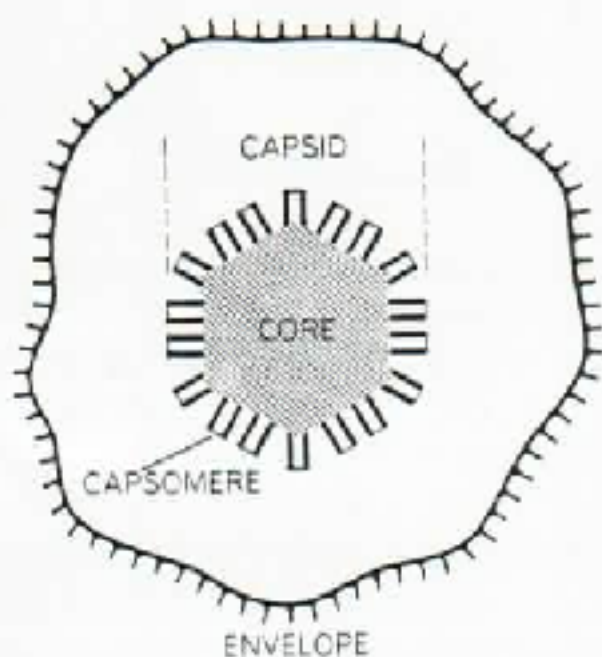
virus particles are enlarged about 58,000 diameters. The shadows indicate that each particle is a regular icosahedron (right). The micrograph was made by Kenneth Smith of the University of Cambridge and Robley C. Williams of the University of California.

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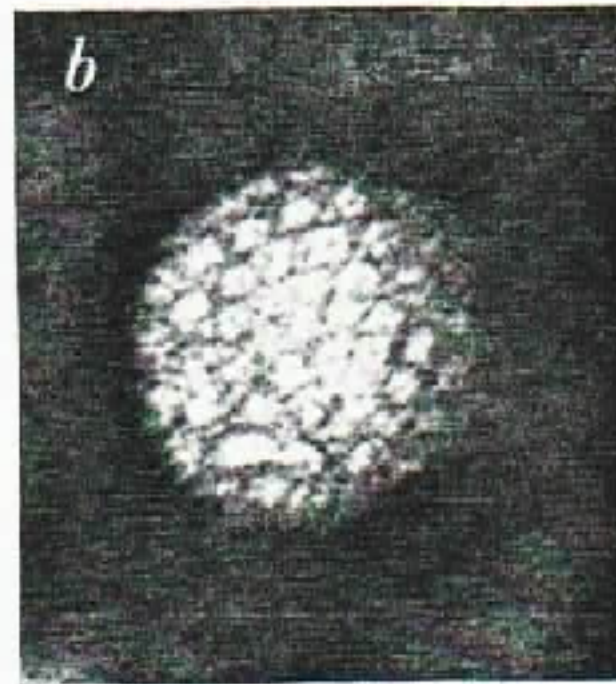
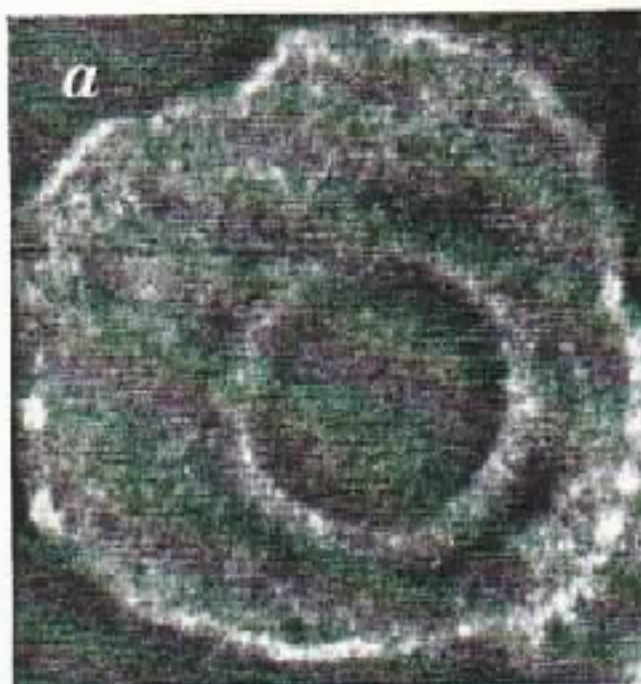


AXES OF SYMMETRY are shown for a regular icosahedron, a figure with 12 corners, 20 faces and 30 edges. Viewed along an axis at any corner, the figure can be rotated in five positions without

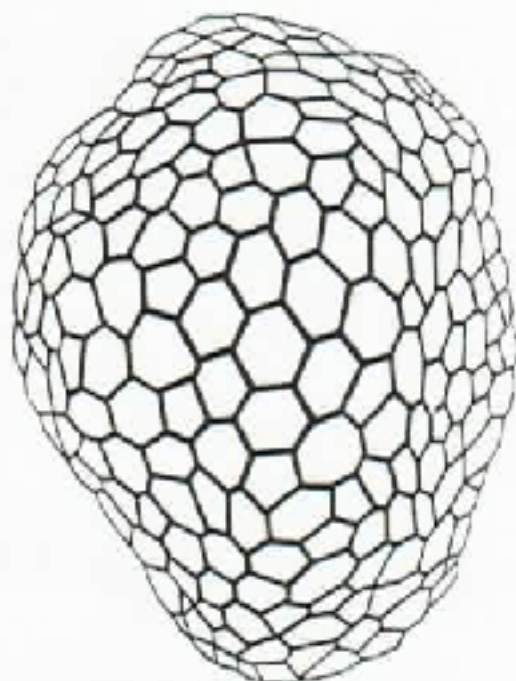
changing its appearance (*left*). Rotated around any face axis, a regular icosahedron exhibits threefold symmetry (*middle*). Rotated around any edge axis, the figure shows twofold symmetry (*right*).



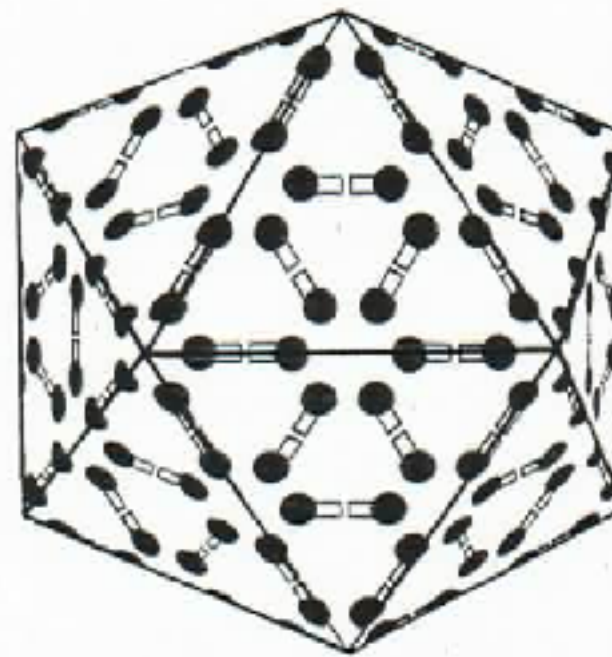
VIRUS NOMENCLATURE covers principal features observed in electron micrographs.



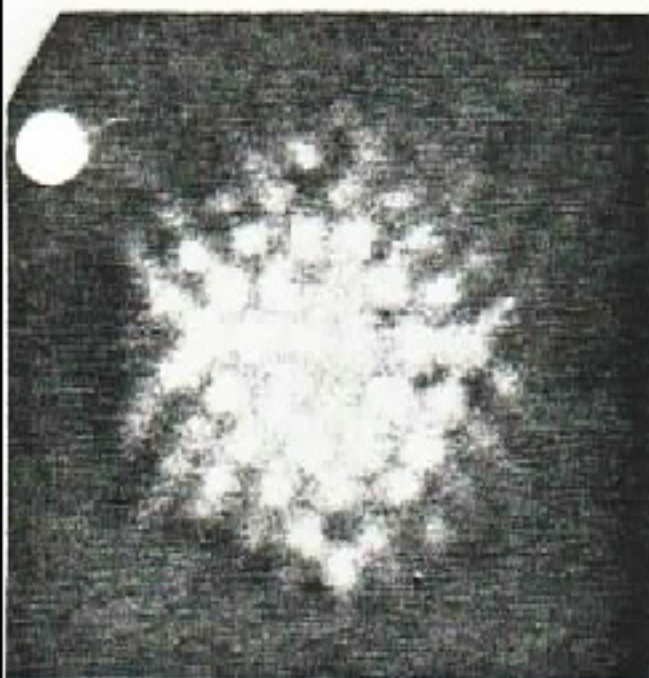
HERPES VIRUS sometimes has an envelope (*a*). The magnification is 310,000 diameters. The capsid (*b*) is composed of 162 capsomeres. Negative staining (*c*) indicates



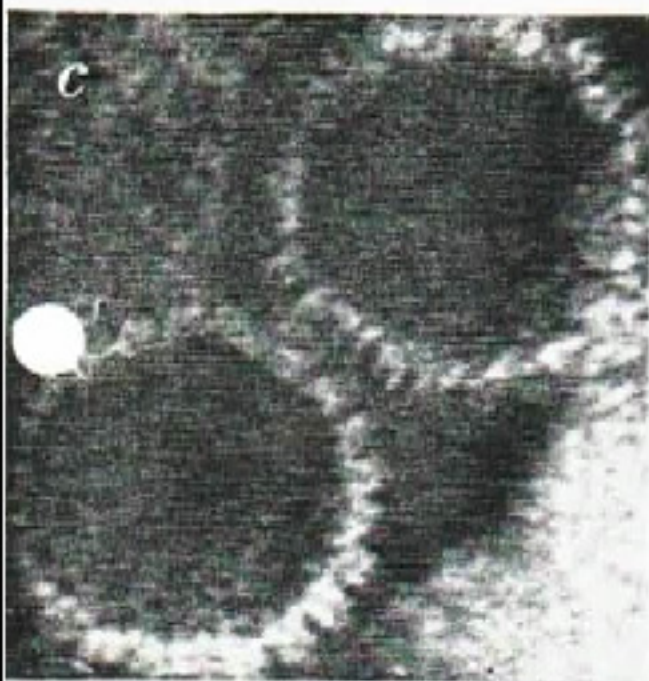
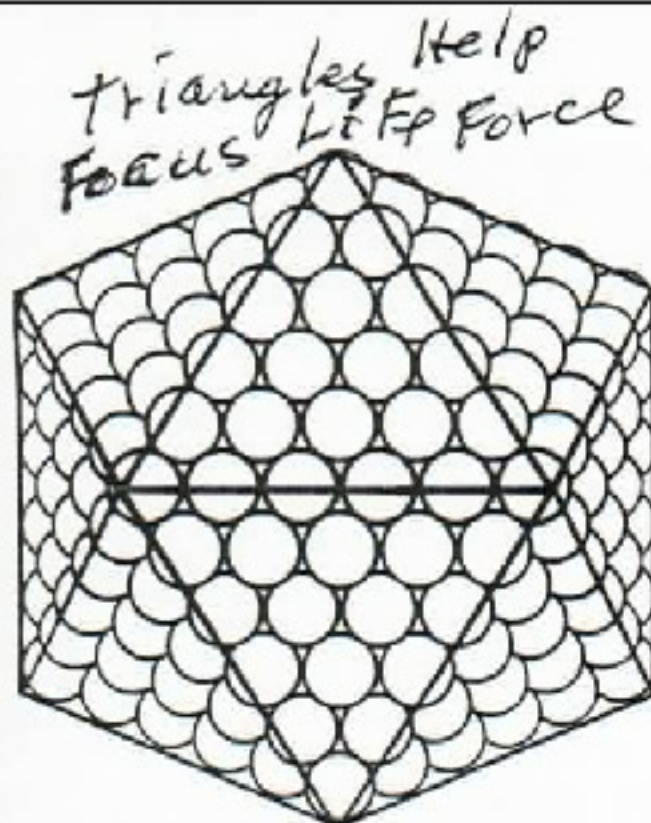
RADIOLARIANS, small marine organisms, have skeletons built of pentagons and hexagons.



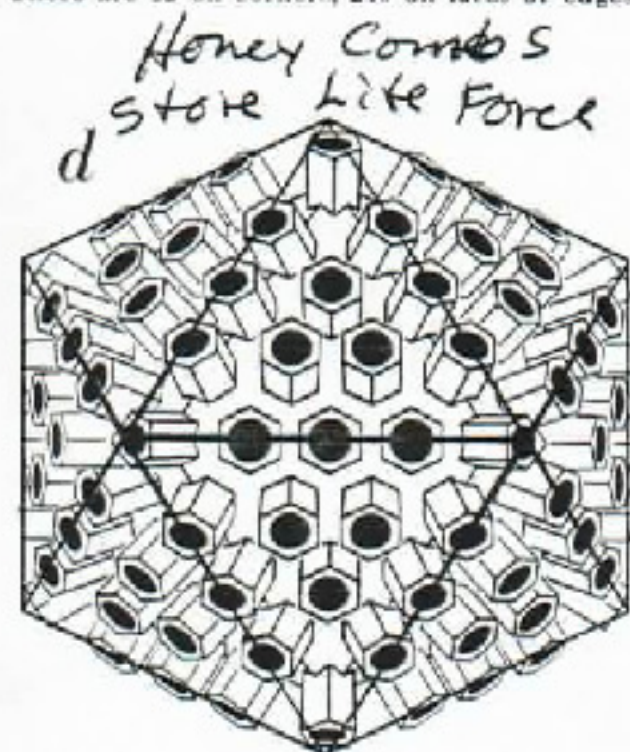
ALTERNATIVE SCHEMES show how a regular icosahedron containing 42 pentagonal and hexagonal capsomeres (*left*) could be built up from 120 (or 240) small subunits.



ADENOVIRUS is shown embedded in phosphotungstate, magnified about one million diameters (left). The drawing shows how the particle's 252 surface subunits, or capsomeres, are arranged with icosahedral symmetry. There are 12 on corners, 240 on faces or edges.



that they are hollow. The drawing (d) shows icosahedral arrangement. Micrographs are by P. Wildy and W. C. Russell of the Institute of Virology in Glasgow and the author.



POLYOMA VIRUS is magnified one million diameters in micrograph by Wildy, M. G. P. Stoker and I. A. Macpherson of the Institute of Virology. It has 42 capsomeres (right).

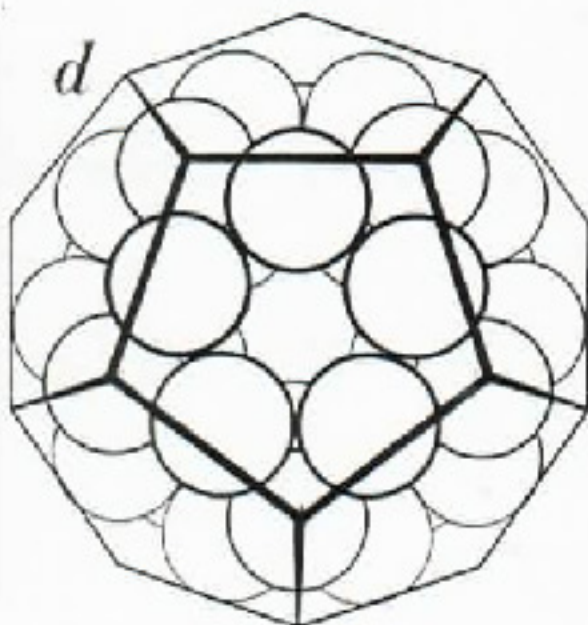
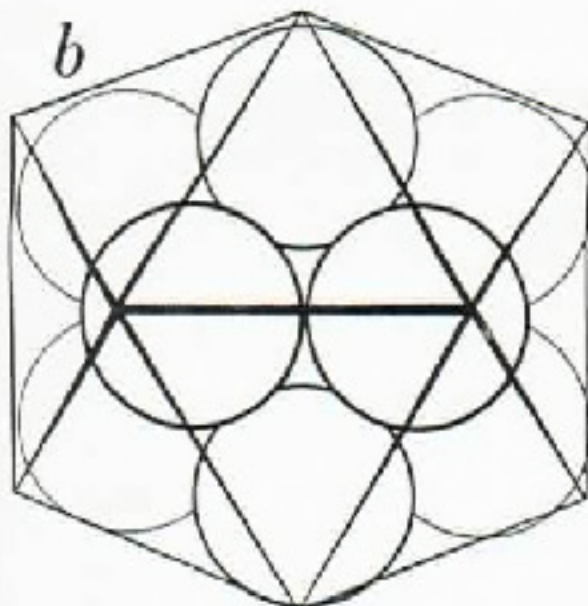
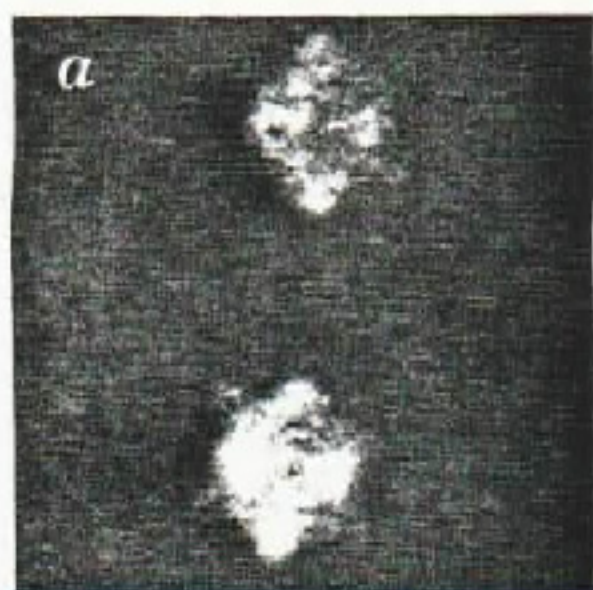


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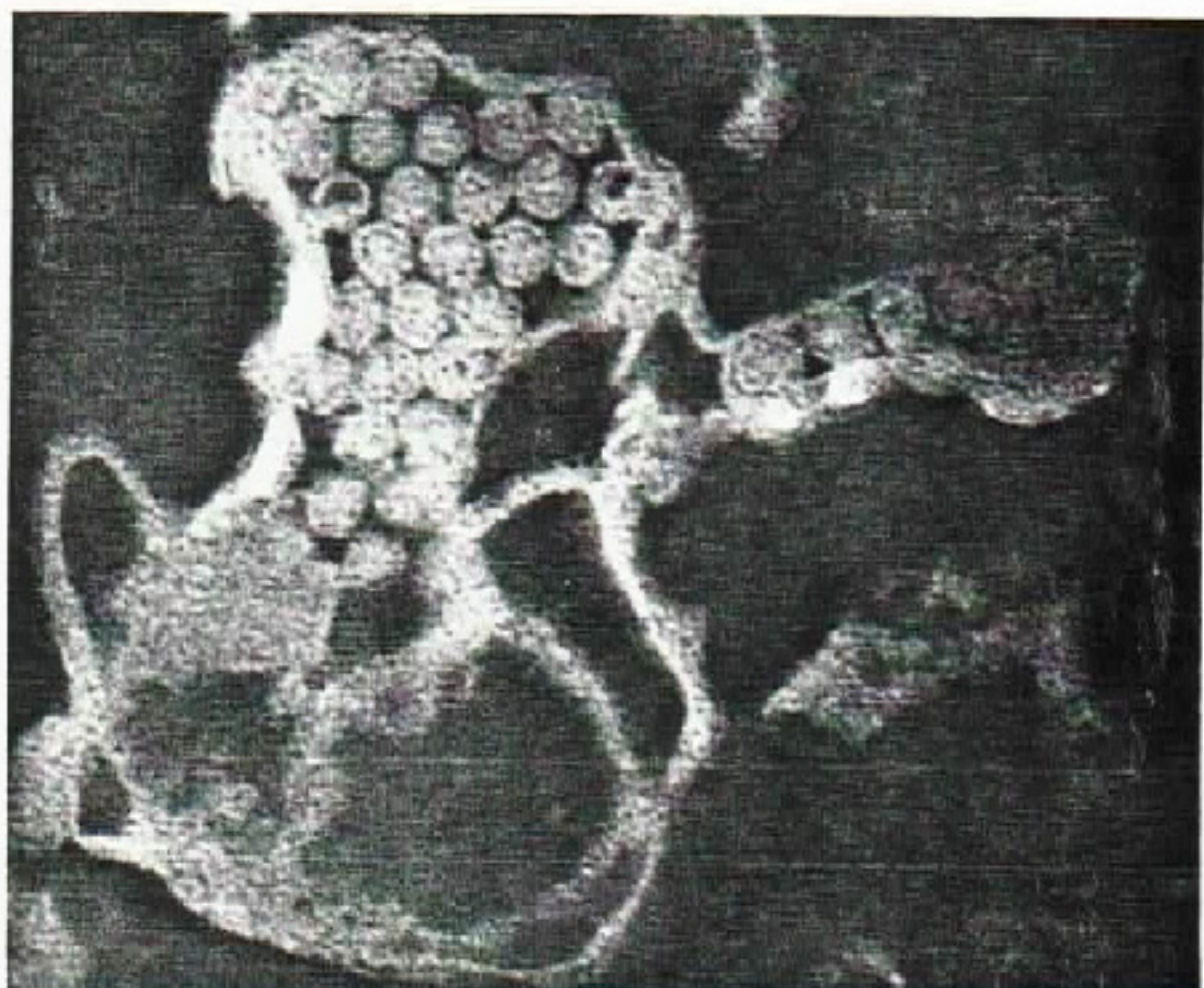
cubic symmetry includes the regular tetrahedron (four faces), dodecahedron (12 faces) and icosahedron (20 faces). Shadowed preparations of the tipula iridescent virus, which causes a disease in the larvae of several insects, showed it to have the shape of a regular icosahedron, and the symmetry was self-evident [see bottom illustration on page 49]. Smaller viruses, on the other hand, do not reveal symmetry unless they are examined at very high magnification, and this requires the use of negative phosphotungstate staining.

Consider the symmetry properties of a regular icosahedron, in which each face is an equilateral triangle. If spokes are projected from the center of the icosahedron through the corners of the triangles, the spokes will represent one axis of rotational symmetry. Spokes projected from the center of the solid through the center of each face will represent a second axis. And spokes projected from the center through the midpoint of each edge will represent a third axis. (There will be 12 corner spokes, 20 face spokes and 30 edge spokes.) If the icosahedron is viewed along the spoke at any corner, one finds that the body can be rotated in five positions without changing its appearance [see top illustration on opposite page]. If the icosahedron is viewed along the spoke at any face, the body can be rotated in three positions without changing its appearance. And if the icosahedron is viewed along an edge spoke, it can be rotated in two positions without change of appearance. The regular icosahedron is thus said to have 5.3.2. symmetry.

Let us see now what implication this symmetry pattern has for a particle of adenovirus, which is associated with respiratory disease in man. The electron microscope shows that the surface of the particle is composed of regularly arranged structural units resembling tiny balls. Moreover, these balls are seen on the vertexes, faces and edges of an icosahedron [see top illustration on this page]. One can identify certain balls surrounded by five neighbors, which indicates that they are located on vertexes and therefore on axes of fivefold symmetry. Balls surrounded by six neighbors must lie on faces or edges and thus must occupy axes of either threefold or twofold symmetry. Along each edge there are six balls, including two balls occupying vertexes. To calculate the total number of balls covering the entire icosahedron one applies the simple formula $10(n - 1)^2 + 2$, where n is the number of balls along one edge. Substituting 6



BACTERIOPHAGE ϕ X174, magnified 750,000 diameters in *a*, appears to consist of 12 capsomeres arranged in icosahedral symmetry as shown in *b*. In other micrographs (*c*) smaller subunits seem to be arranged in ringlike structures. Each capsomere might actually be formed from five subunits as shown in *d*. Thirty such subunits would form a dodecahedron.



POLIOVIRUS PARTICLES are shown inside a fragment of an infected cell. The particles, magnified 250,000 diameters, appear to be composed of subunits smaller than the typical capsomere. The structural arrangement is not established. Electron micrograph is by Jack Nagington of the Public Health Laboratory in Cambridge and the author.

for n yields 252 as the number of morphological units composing the shell of the adenovirus particle.

For purposes of description (and to avoid the term "subunit," which can be applied to morphological, structural or chemical features) I shall adopt the recent terminology suggested for the various viral components [see illustration at middle left on page 50]. The morphological units composing the shell have been given the name "capsomeres." The shell itself is the "capsid." The region inside the capsid is the "core." The outer membrane, seen surrounding the capsid of some viruses, is the "envelope."

One merit of the negative-staining technique is that the electron-dense material is capable of penetrating into extremely small regions between, and even within, the capsomeres. A striking instance of such penetration can be seen in the electron micrograph of the herpes virus shown at the middle left on the preceding page. (In man the herpes virus causes, among other things, "cold sores.") Electron micrographs of the shadowed particle had indicated that it had the same external shape and symmetry as the adenovirus. When the two viruses were negatively stained and still further magnified, however, it could be seen on close examination that the capsomeres of the herpes virus, unlike those of the adenovirus, were elongated hollow prisms, some hexagonal in cross section and others pentagonal. In a number of particles the phosphotungstate penetrated into the central region, or core, normally containing the nucleic acid. In these "empty" particles the elongated capsomeres stand out clearly in profile at the periphery of the virus, and one can see their hollow form and the precision of their radial arrangement.

From the micrographs the number of capsomeres located on each edge was estimated to be five, giving a total of 162 capsomeres for the herpes virus. Of the 162 capsomeres, 12 are pentagonal prisms and 150 are hexagonal prisms. To satisfy the packing arrangement in accordance with icosahedral symmetry, the 12 pentagonal prisms would have to be placed at the corners and the 150 hexagonal prisms located on the edges or faces of the particle [see drawing at middle right on preceding page].

The need for pentagonal units goes deeper than the simple need to satisfy icosahedral symmetry. As early geometers observed, there is no way to arrange a system of hexagons so that they will enclose space. But if pentagonal units are included with hexagons, it is possible to enclose space in an almost